Practice School II Report (Mid semester), 2026

Learning Management System System @Eurobliz

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CERTIFICATE

This is to certify that the Practice School II internship report titled **"Software Development Internship at Eurobliz Technologies"** submitted by **Kushagra Gupta (2022BTECH051)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** at **Institute of Engineering and Technology, JK Lakshmipat University, Jaipur**, is a record of the work carried out by him during the internship period from **15 January 2026 to 15 June 2026**.

The student has worked as a **Software Developer Intern** at **Eurobliz Technologies**, Lille, France, under the guidance of **Mr. Soumyadip Saha (External Supervisor)** and **Dr. S. Taruna (Faculty Supervisor)**.

This report represents the work completed by the student during the mid-term phase of the Practice School II internship and has been prepared as per the academic requirements of the university.

02/03/2026

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CONTENTS

Title	Page No.
CERTIFICATE	1
ACKNOWLEDGEMENT	2
ABSTRACT	4
LIST OF FIGURES	5
Chapter 1: ABOUT THE ORGANIZATION	6
Chapter 2: INTRODUCTION	7
Chapter 3: TOOLS & TECH STACK USED	9
Chapter 4: ROLES AND RESPONSIBILITIES	11
Chapter 5: WORK DONE	14
Chapter 6: FUTURE PLAN	23
Chapter 7: CONCLUSION	25
REFERENCES	26

ABSTRACT

This report describes the work carried out during the Practice School II internship at Eurobliz Technologies, where I worked as a Software Developer Intern. The internship focused on gaining practical experience in full-stack software development and understanding how modern, scalable web applications are designed, developed, and maintained in a professional environment.

During the internship, I contributed primarily to the development of a multi-tenant Learning Management System (LMS) designed for educational institutions. The system supports multiple schools with secure data isolation, role-based access control, and subscription-based feature management. My work included designing database schemas using PostgreSQL and SQLAlchemy, developing backend APIs using FastAPI, implementing secure authentication using JWT, and building frontend dashboards using React and TypeScript. I also integrated cloud storage services such as MinIO and Cloudflare R2, and implemented system optimization using Redis caching and background job processing.

In addition to the LMS project, I contributed to the PlayEvent client platform by developing backend APIs, designing database schemas, and integrating frontend and backend systems. I also explored modern backend technologies such as Bun, Hono, and Drizzle ORM, and studied modern system architecture, authentication workflows, and real-time backend systems.

The internship provided valuable hands-on experience in database design, backend development, frontend integration, authentication systems, and scalable software architecture. It helped me understand real-world software development workflows, including system design, testing, debugging, and performance optimization. This report presents the organization overview, tools and technologies used, roles and responsibilities, work completed so far, and future plans for the internship.

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 1	LMS Database Schema	14
Figure 2	LMS Dashboard Interface	15
Figure 3	MinIO Object Storage Dashboard	16
Figure 4	Love Passport application interface	18
Figure 5	Model Context Protocol (MCP) architecture	19
Figure 6	Backend API testing	20
Figure 7	GitHub contribution graph	20

Chapter 1: ABOUT THE ORGANIZATION

1.1 GOALS & MISSIONS

Eurobliz Technologies is a digital technology company based in Lille, France. The company focuses on helping startups and businesses build modern and scalable digital products. Eurobliz works as a technology partner and supports clients in developing reliable software systems and cloud-based solutions.

The main goal of Eurobliz is to provide secure, scalable, and efficient technology solutions that help businesses grow. The company also provides technical consulting to help startups choose the right technologies and system architecture.

Eurobliz supports innovation and entrepreneurship by helping startups and student entrepreneurs develop digital platforms and software solutions.

1.2 SERVICES & PRODUCTS OFFERED

Eurobliz Technologies provides software development and technology consulting services. The main services include:

- Website development for businesses and startups
- Hybrid mobile application development
- Cloud-based solutions and system integration
- Internet of Things (IoT) system development
- Search Engine Optimization (SEO) services
- Analytics and business intelligence solutions

Eurobliz develops scalable web applications, backend systems, and cloud solutions based on client requirements.

Chapter 2: INTRODUCTION

2.1 Project Overview

During my Practice School II internship at Eurobliz Technologies, I worked as a Software Developer Intern. The main focus of my internship was to contribute to the development of real-world software applications and understand how modern full-stack systems are designed and implemented.

My primary work involved contributing to the development of a Learning Management System (LMS) for a client. The LMS is a web-based platform used to manage educational activities such as course creation, student enrollment, assignment submission, grading, and user management. The system supports multiple user roles such as Admin, Principal, Teacher, and Student, each with specific permissions and responsibilities.

In addition to the LMS project, I also worked on backend and frontend development tasks as practice before working on other client applications. My work involved designing database schemas, developing backend APIs, implementing authentication systems, building frontend components, and integrating different system modules. After gaining initial experience with similar technologies, I contributed to the development of client applications.

The internship provided hands-on experience in working with real software systems, understanding project requirements, and building scalable and maintainable applications using modern development tools and technologies.

2.2 Problem Definition

Modern web applications require proper system design to ensure scalability, security, and maintainability. Without a structured architecture, software systems become difficult to manage, extend, and secure.

During the internship, several key challenges were identified in developing enterprise applications such as the Learning Management System:

- **Designing database** schemas that efficiently store and manage large amounts of data
- Developing structured **backend APIs** that are secure and easy to maintain
- Implementing **authentication systems** to ensure secure user access
- Managing role-based access control **(RBAC)** for different types of users
- Integrating frontend and backend systems to ensure smooth communication
- Building reusable frontend components to improve development efficiency
- Handling **file storage** securely using cloud storage systems
- Ensuring system performance and reliability through optimization and testing

To solve these problems, modern technologies and structured development practices were used to design and build scalable and secure applications.

2.3 Background Study and Learning

Before starting the internship, I had academic knowledge in programming, database management, and software development. This helped me understand the project requirements and technologies used during the internship.

During the internship, I gained deeper knowledge in the following areas:

Full-Stack Application Architecture
I learned how frontend, backend, and database systems work together in a real-world application. I understood how data flows between different system components.

Database Design and Management
I learned how to design scalable database schemas using normalization and relationship mapping. I implemented database models and managed database migrations.

Backend Development
I developed backend APIs using FastAPI. I implemented CRUD operations, authentication systems, and role-based access control.

Frontend Development
I built frontend components using React and integrated them with backend APIs. I learned how to manage application state and handle form validation.

Authentication and Security
I implemented secure authentication systems using JWT tokens and role-based access control to ensure secure user access.

Cloud Storage and System Integration
I worked with cloud storage systems such as MinIO and Cloudflare R2 to manage file storage securely.

Testing and Debugging
I learned different testing methods and tools to ensure system reliability and correct functionality.

This internship helped me understand real software development workflows and improved my practical skills in building modern web applications.

Chapter 3: TOOLS & TECH STACK USED

During the internship, modern tools and technologies were used for backend development, frontend development, database management, storage integration, testing, and system architecture. These tools helped in building scalable, secure, and maintainable applications.

Programming Languages

- **Python –** Backend development using FastAPI, API logic, authentication, and database operations
- **JavaScript –** Frontend development using React
- **TypeScript –** Type-safe frontend development and API integration

Backend Technologies and Architecture

- **FastAPI –** REST API development and backend system implementation
- **SQLAlchemy –** ORM for database interaction and model definition
- **Alembic –** Database migration and schema version control
- **JWT Authentication –** Secure authentication using access and refresh tokens
- **Argon2 –** Secure password hashing and credential protection
- **Role-Based Access Control (RBAC) –** Access control based on user roles
- **Pydantic –** Request and response validation
- **Async FastAPI –** Asynchronous backend execution for performance improvement
- **Controller–Service Architecture –** Separation of business logic and API routes
- **Soft Delete Logic –** Safe deletion and recovery of database records

Database Technologies

- **PostgreSQL –** Primary relational database system
- Database normalization and relationship design
- Schema design with constraints and scalability planning
- ACID-compliant database transactions

Frontend Technologies and Libraries

- **React –** User interface development using reusable components
- **TanStack Query –** API data fetching, caching, and state synchronization
- **TanStack Router –** Application routing and navigation
- **React Hook Form –** Form handling and validation
- **Zod –** Data validation
- **Zustand –** Global state management
- **Axios –** API communication and HTTP requests
- **Tailwind CSS –** Responsive UI design
- **TypeScript –** Type-safe frontend–backend integration

Storage and Cloud Technologies

- **MinIO –** Object storage for file management
- **Cloudflare R2 –** Cloud-based object storage
- **Pre-signed URLs –** Secure file upload and download

- Cloud storage integration for secure external file handling

Testing and Quality Assurance Tools

- **Pytest –** Backend testing
- **Swagger UI –** API testing and validation
- **Vitest and Jest –** Frontend testing tools (studied and explored)
- **Playwright and Cypress –** End-to-end testing tools
- Manual testing and debugging for system validation

Development Tools and System Technologies

- **Git and GitHub –** Version control and collaboration
- **Docker –** Containerized database and development environment
- **Redis –** Caching for performance optimization
- **APScheduler –** Background job scheduling and automation
- **Convex –** Modern backend platform exploration
- **Better Auth –** Authentication framework exploration
- **MCP (Model Context Protocol) –** Backend communication protocol study

Modern Backend and Framework Technologies Explored

- **Bun –** Modern JavaScript runtime for backend development
- **Hono –** Lightweight backend framework used with Bun
- **Drizzle ORM –** Type-safe ORM used with PostgreSQL
- Modern backend architecture and authentication implementation

AI-Assisted Development and Productivity Tools

- AI coding tools for debugging, code generation, and learning
- VS Code and modern development environment tools
- AI-assisted development for improving productivity

Chapter 4: ROLES AND RESPONSIBILITIES

4.1 Objectives

The main objective of my internship at Eurobliz Technologies was to gain practical experience in real-world software development by working on production-level applications. The internship focused on understanding full-stack system architecture, backend and frontend development, database design, authentication systems, and modern development practices.

The key objectives were:

- To understand enterprise application architecture and development workflow
- To design scalable and structured database schemas
- To develop secure backend APIs and authentication systems
- To build modular and reusable frontend components
- To integrate frontend, backend, and storage systems
- To learn testing, debugging, and system optimization techniques
- To work on real client projects and production-level systems

4.1.1 Learning Objectives

During the internship, I focused on learning and implementing:

- Backend development using FastAPI, SQLAlchemy, and PostgreSQL
- Frontend development using React, TypeScript, and modern libraries
- Authentication systems using JWT, refresh tokens, and role-based access
- Database schema design, normalization, and migrations
- API integration, validation, and structured architecture
- Cloud storage integration using MinIO and Cloudflare R2
- Testing, debugging, and performance optimization
- Modern backend technologies such as Bun, Hono, and Drizzle ORM

These objectives helped me gain hands-on experience in building scalable and secure applications.

4.2 ROLES AND RESPONSIBILITIES

During my internship at Eurobliz Technologies, I worked as a Software Developer Intern and was responsible for contributing to backend development, frontend development, database design, authentication systems, and system integration across internal projects and client applications. My role involved developing scalable system components, implementing secure authentication, and supporting production-level software development.

My responsibilities across different projects are summarized below:

Learning Management System (LMS)

Role: Full-Stack Developer Intern

Responsibilities:

- Designed and implemented database schema and relationships using PostgreSQL and SQLAlchemy
- Developed backend APIs and authentication systems using FastAPI and JWT
- Built frontend dashboards and integrated APIs using React and TypeScript
- Implemented multi-tenant architecture, role hierarchy, and access control
- Integrated file storage systems and supported performance optimization

PlayEvent Platform (Client Project)

Role: Backend and Frontend Developer Intern

Responsibilities:

- Designed database schema and implemented backend APIs
- Implemented authentication, access control, and business logic
- Integrated frontend with backend APIs and updated frontend types
- Tested APIs and ensured proper system functionality

Modern Backend Development (bun-hono-drizzle-auth)

Role: Backend Developer Intern (Learning and Implementation)

Responsibilities:

- Developed backend APIs using Bun and Hono framework
- Implemented authentication workflows using Better Auth
- Managed database schema and migrations using Drizzle ORM
- Studied modern backend architecture and authentication systems

Modern React Application Architecture Projects

Role: Frontend Developer Intern

Responsibilities:

- Developed reusable frontend components and form systems
- Implemented API integration and state management using modern React tools
- Structured frontend architecture for scalability and maintainability

Convex-Based Applications (Love Passport Project)

Role: Full-Stack Developer Intern (Learning and Implementation)

Responsibilities:

- Developed frontend using React Native and TypeScript
- Implemented backend logic using Convex platform
- Worked with real-time data synchronization and authentication systems

MCP Server and TanStack Form Integration

Role: Backend Developer Intern (Exploration and Implementation)

Responsibilities:

- Implemented MCP server integration for backend communication
- Studied secure interaction between backend systems and AI tools
- Explored modern backend communication protocols

General Responsibilities Across Projects

In addition to project-specific work, my overall responsibilities included:

- Developing backend APIs and database models
- Implementing authentication and access control systems
- Integrating frontend, backend, and storage systems
- Testing APIs and debugging system issues
- Supporting system optimization and performance improvements
- Working with Git and following professional development practices

Chapter 5: WORK DONE

5.1 Internship Work Done So Far

During my internship at Eurobliz Technologies, I worked on multiple internal systems and real client applications, primarily focusing on full-stack development, database design, backend API development, frontend implementation, authentication systems, and scalable system architecture.

The major projects I contributed to are described below.

5.1.1 Learning Management System (LMS)

The main project during my internship was the development of a multi-tenant Learning Management System (LMS) designed for educational institutions. The system supports multiple schools with strict data isolation, role-based access control, and subscription-based feature management.

<u>Database Design</u>

I designed and implemented the database schema using PostgreSQL and SQLAlchemy. This included creating tables for users, schools, courses, assignments, submissions, and roles. I defined relationships using primary and foreign keys and applied normalization to ensure efficient data storage.

I also implemented database migrations using Alembic and added features such as soft delete, restore functionality, and structured relationship mapping.

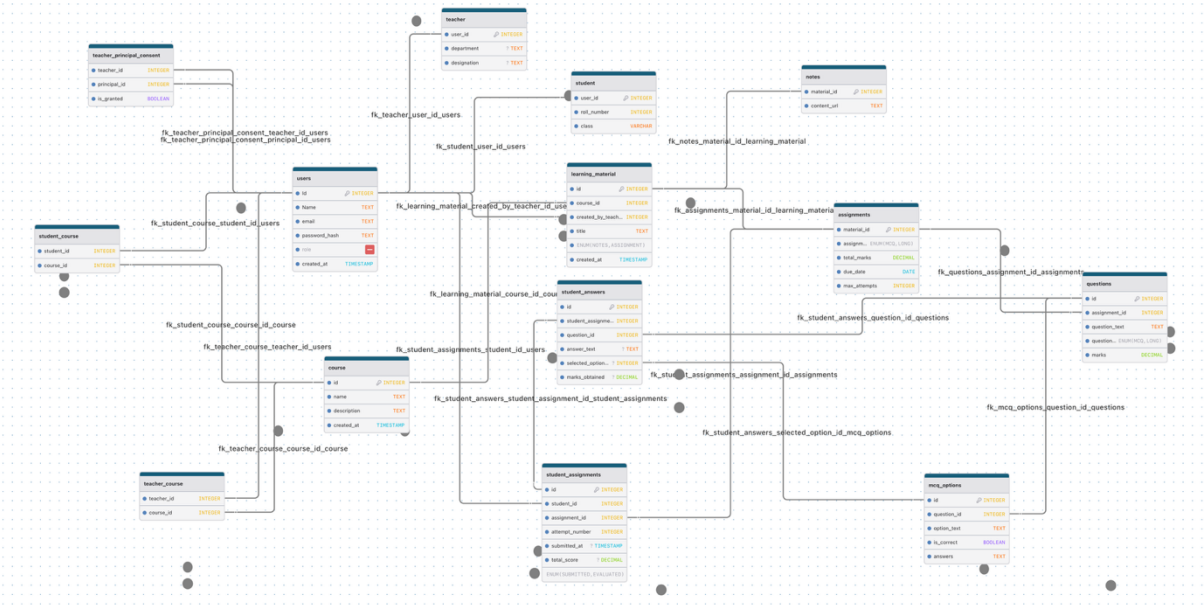

Figure 1: Database Schema of the Learning Management System showing relationships.

<u>Backend Development</u>

I developed backend APIs using FastAPI. This included implementing CRUD operations, authentication, and structured API routes. I implemented secure authentication using JWT

access tokens and refresh tokens, along with session management and role-based access control.

The backend followed controller–service architecture, which improved code maintainability and separation of concerns. I also converted backend operations to asynchronous execution to improve performance and concurrency handling.

Frontend Development

I developed frontend components using React and TypeScript. I implemented dashboards for different user roles such as Admin, Principal, Teacher, and Student.

I used TanStack Query for API integration, Zustand for state management, and React Hook Form with Zod for form handling and validation. I implemented authentication flows, protected routes, and reusable UI components.

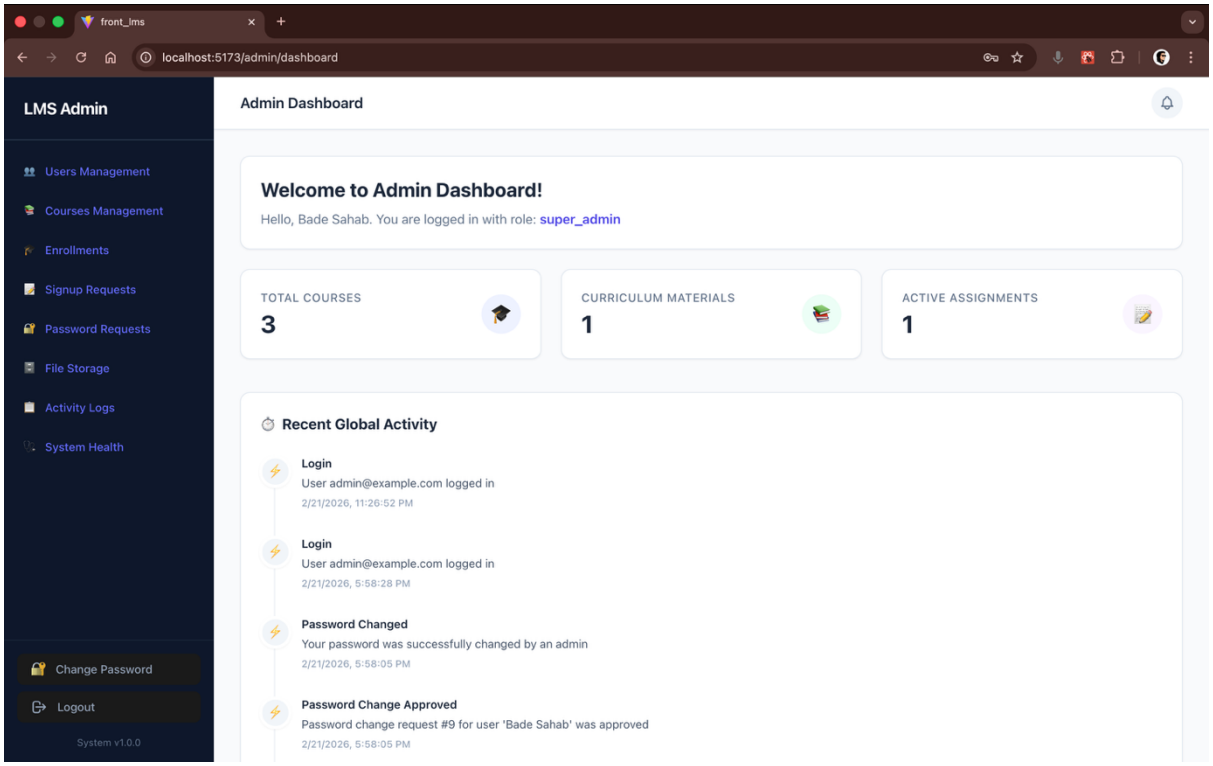

Figure 2: LMS Dashboard Interface showing role-based user interface and course management system.

Multi-Tenant Architecture

I implemented a multi-tenant system that supports multiple schools within the same platform. Each school has isolated data, role hierarchy, and subscription-based access control.

I created a school entity and implemented hierarchical role structure such as Admin → Principal → Teacher → Student. This ensured secure and structured access control across the system.

Storage Integration and System Optimization

I integrated MinIO and Cloudflare R2 for secure file storage. I implemented file upload and download using pre-signed URLs.

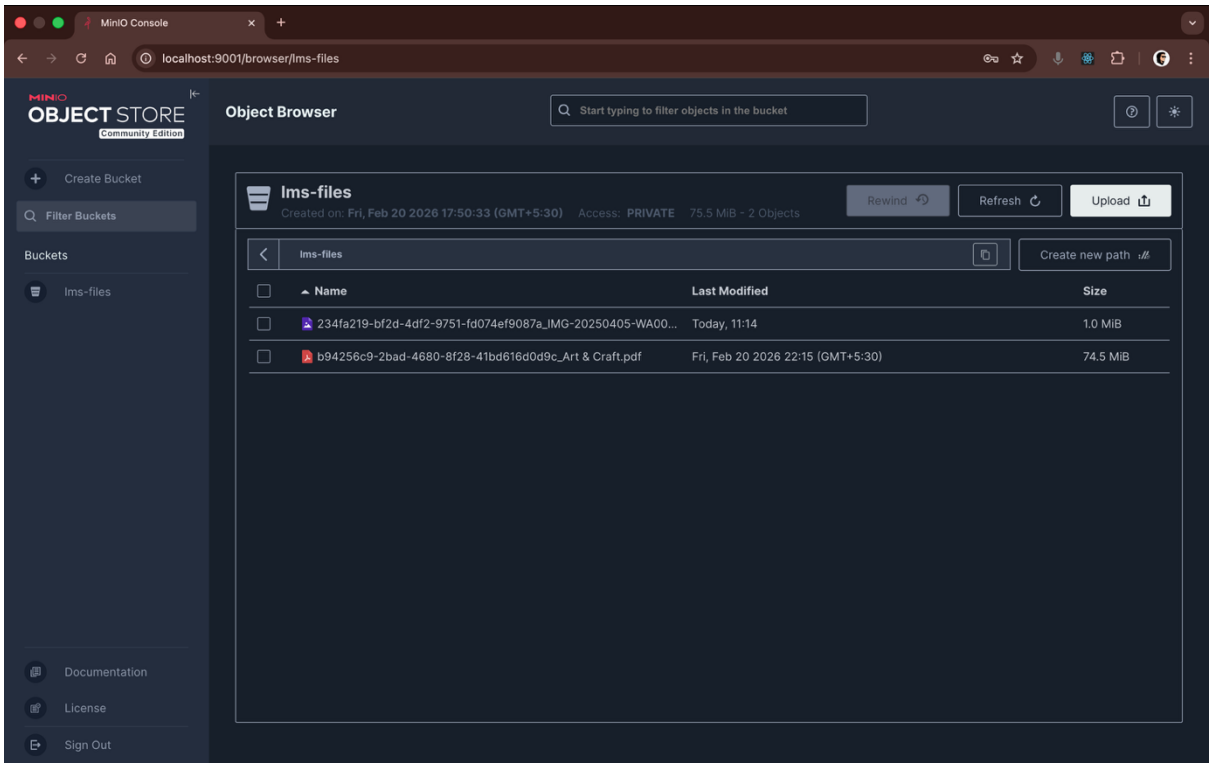

Figure 3: MinIO Object Storage Dashboard used for secure file storage in the LMS system.

I also implemented Redis caching to improve backend performance and reduce database load. Background jobs were implemented using APScheduler for handling automated system tasks.

This project provided complete experience in building a real production-level full-stack system.

5.1.2 PlayEvent Platform (Client Project)

I contributed to backend and frontend development for the PlayEvent platform, which is a client application used by European football organizations.

<u>Database and Backend Development</u>

I designed database schemas and implemented backend APIs using FastAPI and SQLAlchemy. I created multiple database tables, defined relationships, and implemented structured API routes.

I implemented authentication, authorization, and role-based access control. I also tested APIs using Swagger UI and verified system functionality.

<u>Frontend Integration</u>

I worked on frontend updates to align the frontend with backend APIs. I updated TypeScript types, integrated APIs using TanStack Query, and ensured proper data flow between frontend and backend systems.

This project helped me gain experience working on real client systems and production-level code.

5.1.3 Modern Backend Development using Bun, Hono, and Drizzle

I developed and experimented with a modern backend system using Bun runtime, Hono framework, and Drizzle ORM.

I implemented authentication workflows using Better Auth and managed database schema using Drizzle ORM. I developed backend APIs and learned modern backend architecture and scalable authentication systems.

This helped me understand alternative backend technologies and modern backend development approaches.

5.1.4 Modern React Application Architecture

I worked on modern React-based applications to understand scalable frontend architecture and reusable component design.

I developed reusable form components using React Hook Form and Zod. I implemented API integration using Axios and TanStack Query and used Zustand for global state management.

I implemented modular frontend architecture with proper folder structure, reusable components, and service-based API integration.

This improved my understanding of scalable frontend system design.

5.1.5 Convex-Based Application and Real-Time Systems

I worked on experimental applications using Convex backend platform, including the Love Passport project.

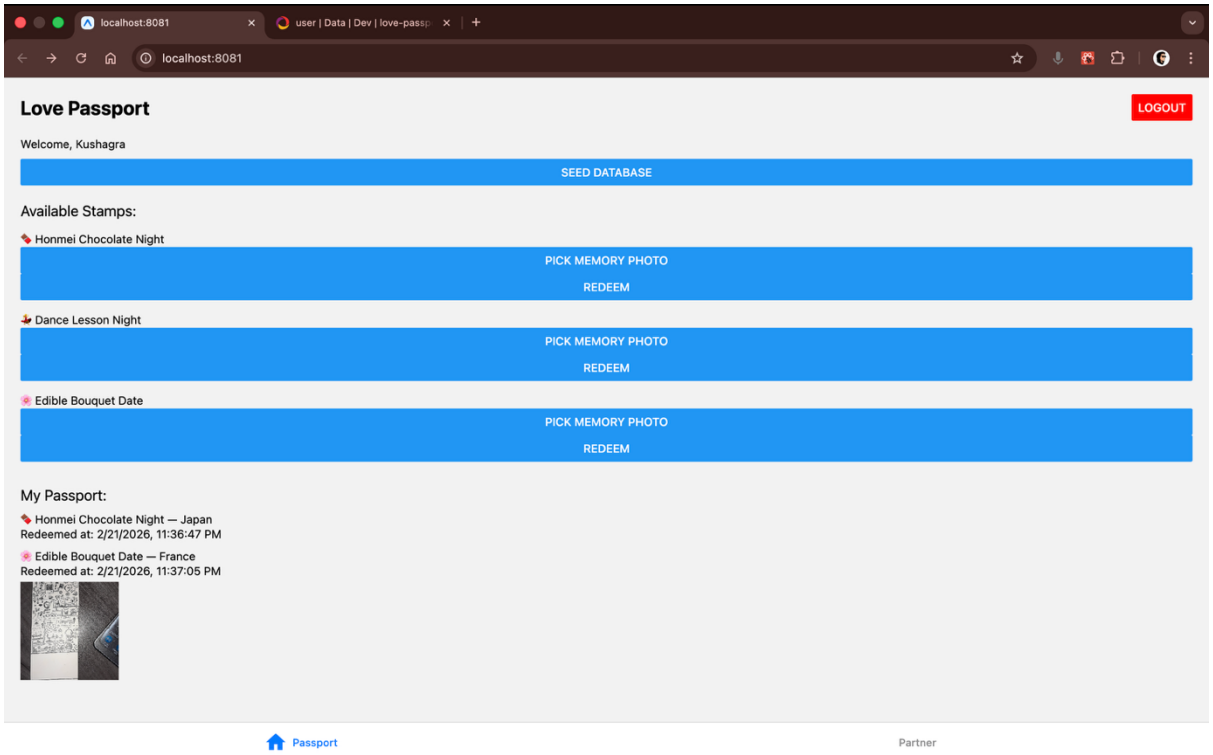

Figure 4: Love Passport application interface demonstrating real-time backend integration using Convex.

I implemented frontend using React Native and TypeScript and integrated backend using Convex database and sync engine. This allowed real-time data synchronization and modern backend communication.

I also explored authentication workflows using Better Auth and learned modern backend architecture concepts.

5.1.6 MCP Server and AI Integration

I worked on Model Context Protocol (MCP) based systems to understand AI-backend integration.

I implemented MCP server integration and studied how backend systems can securely interact with AI tools and external systems.

This provided exposure to modern backend communication protocols and AI-assisted development systems.

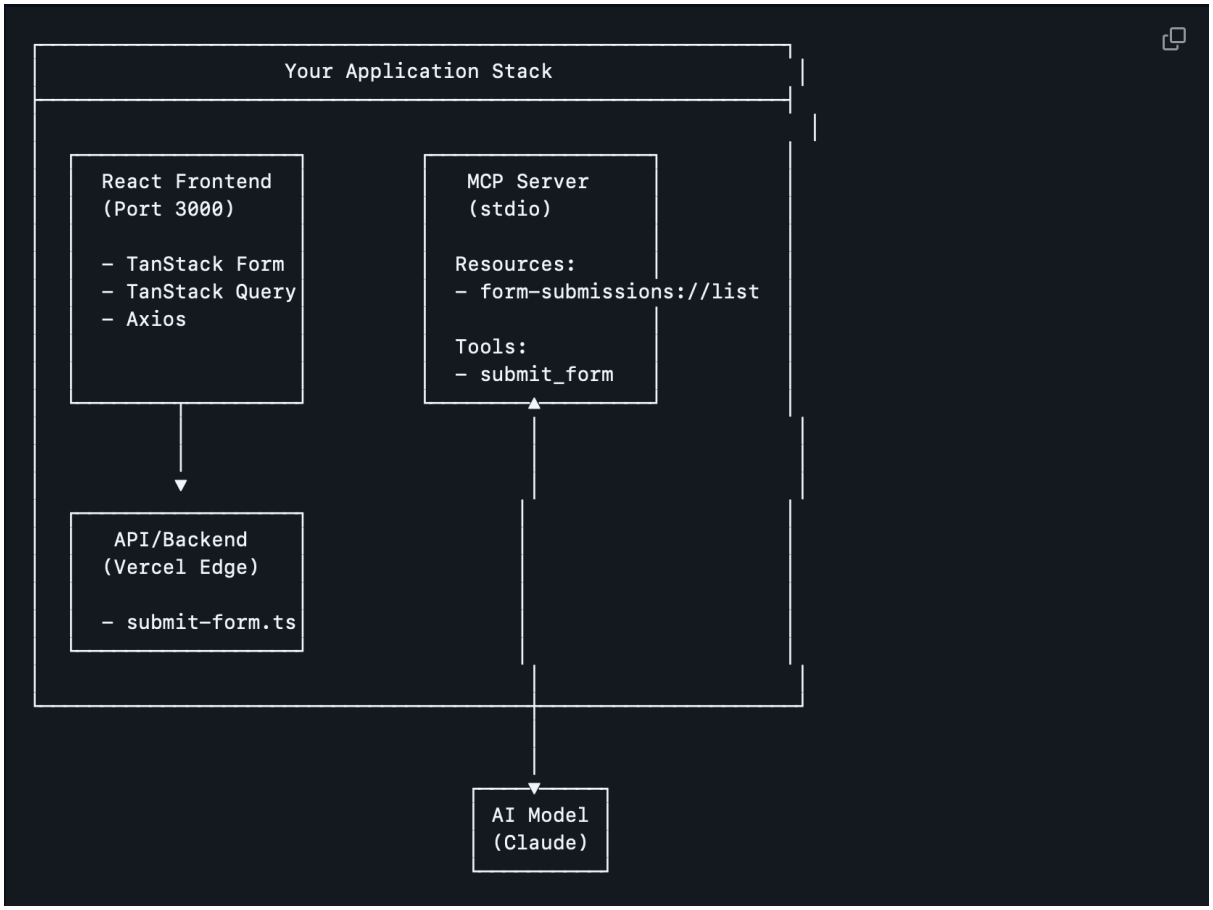

Figure 5: Model Context Protocol (MCP) architecture showing interaction between AI systems and backend services.

5.1.7 System Integration, Testing, and Optimization

In addition to project development, I worked on system integration, testing, and performance optimization.

My work included:

- Testing backend APIs using Swagger UI
- Debugging backend and frontend issues
- Refactoring code to improve maintainability
- Implementing Redis caching for performance improvement
- Implementing background jobs using APScheduler
- Improving system architecture and reliability

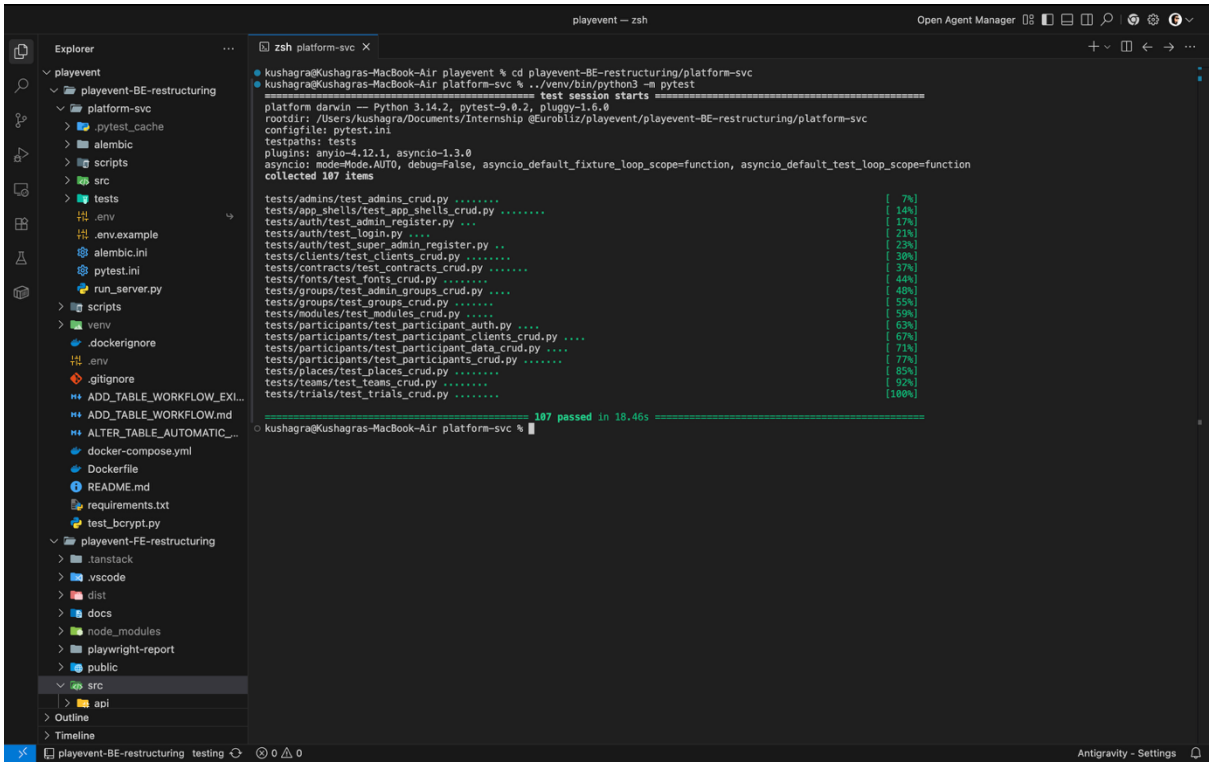

Figure 6: Backend API testing using Swagger and terminal verification.

All development work was managed using Git and GitHub for version control and collaborative development.

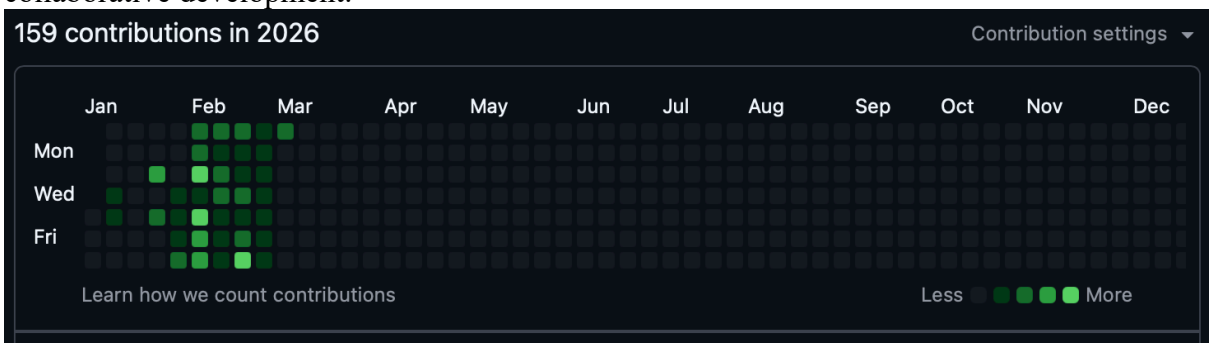

Figure 7: GitHub contribution graph showing active development during the internship period.

Through these projects and responsibilities, I gained practical experience in full-stack development, backend architecture, database design, authentication systems, frontend development, and modern backend technologies while working on real production-level systems.

5.2 Week-wise Log

This section summarizes the major work performed during the internship on a weekly basis.

Week 1: Orientation and System Setup

- Completed onboarding and understood company workflow and development practices

- Studied LMS architecture, database design concepts, and API communication
- Set up development environment with required tools, database, and backend framework

Week 2: Database Design and Backend Foundation

- Designed LMS database schema with entities, relationships, and normalization
- Implemented database models using SQLAlchemy and migrations using Alembic
- Set up FastAPI backend project and connected it with PostgreSQL

Week 3: Backend API Development and Authentication

- Developed REST APIs and implemented CRUD operations
- Implemented secure authentication using JWT access and refresh tokens
- Added role-based access control, session management, and password security

Week 4: Backend Optimization and Frontend Initialization

- Converted backend APIs to asynchronous execution for better performance
- Set up frontend project using React, TypeScript, and Tailwind CSS
- Studied modern frontend tools such as TanStack Query, Zustand, and Axios

Week 5: Frontend Development and Integration

- Developed reusable frontend components and forms using React Hook Form and Zod
- Integrated frontend with backend APIs using Axios and TanStack Query
- Structured frontend architecture with modular and scalable design

Week 6: Storage Integration and Modern Backend Exploration

- Integrated MinIO and Cloudflare R2 for secure file storage
- Implemented file upload and access using pre-signed URLs
- Explored Convex backend platform and MCP server integration

Week 7: Client Project (PlayEvent) Backend and Frontend Development

- Designed database schemas and developed backend APIs for client application
- Implemented authentication, role hierarchy, and access control logic
- Integrated frontend with backend APIs and validated system functionality

Week 8: LMS Frontend Development and System Integration

- Developed LMS dashboards and role-based user interfaces
- Implemented authentication flows, protected routes, and state management
- Integrated frontend, backend, and storage systems

Week 9: System Optimization and Advanced Backend Development

- Implemented Redis caching to improve backend performance

- Integrated background jobs using APScheduler
- Explored modern backend technologies including Bun, Hono, Drizzle, and Better Auth

Week 10: Multi-Tenant Architecture and Final System Integration

- Implemented multi-tenant LMS architecture with school-level data isolation
- Developed hierarchical role structure and subscription-based access control
- Completed system integration, testing, and architecture improvements

This weekly progress reflects continuous learning and active contribution to real-world software development, including full-stack development, client project implementation, and modern backend and frontend system architecture.

Chapter 6. FUTURE PLAN

The internship will continue until June 2026, and the future work will focus on improving existing systems, contributing to client projects, and learning modern backend, frontend, and AI technologies. The main future plans are described below.

6.1 Learning Management System (LMS) Improvements

I will continue working on improving the Learning Management System by adding new features, improving performance, and enhancing system scalability. This includes improving multi-tenant architecture, optimizing backend performance, enhancing frontend user experience, and strengthening authentication and access control.

I will also work on improving the system architecture to make the LMS more scalable and production-ready.

6.2 Modern Backend Development using Bun, Hono, and Drizzle

I will continue working on the bun-hono-drizzle-auth project to gain deeper understanding of modern backend development. This includes implementing scalable authentication systems, improving backend architecture, and understanding real-world usage of modern backend frameworks.

This will help me learn alternative backend technologies and scalable backend system design.

6.3 Client Project Contribution (PlayEvent Platform)

After gaining deeper understanding of system architecture and development practices, I will continue contributing to the PlayEvent client platform. This includes implementing new backend features, improving existing APIs, and ensuring proper frontend-backend integration.

This will help me gain more experience working on real production-level client systems.

6.4 AI Integration and AI Application Development

I will focus on learning and implementing AI integrations in web applications. This includes integrating AI features into applications, understanding AI-assisted development workflows, and building AI-powered systems.

This will help me understand modern AI application development and its integration with backend and frontend systems.

6.5 SaaS Deployment and Scalability

I will work on improving the LMS system to make it a fully scalable Software-as-a-Service (SaaS) platform. This includes learning deployment processes, improving system scalability, optimizing performance, and preparing the system for real-world production environments.

I will also learn how modern applications are deployed and managed at scale using proper backend architecture, database optimization, and cloud deployment practices.

6.6 Continuous Learning and Technology Exploration

I will continue learning new technologies, tools, and development practices to improve my technical skills. This includes exploring modern backend frameworks, frontend tools, cloud technologies, and system architecture concepts.

This continuous learning will help me become a better software developer and contribute effectively to real-world software projects.

Chapter 7. CONCLUSION

The Practice School II internship at Eurobliz Technologies provided valuable hands-on experience in real-world software development. During this internship, I worked as a Software Developer Intern and contributed to the development of a multi-tenant Learning Management System (LMS), client applications such as the PlayEvent platform, and several modern backend and frontend systems.

Through the LMS project, I gained complete experience in full-stack development, including database schema design, backend API development using FastAPI, frontend implementation using React and TypeScript, authentication systems, and cloud storage integration. I also implemented advanced features such as multi-tenant architecture, role-based access control, Redis caching, and background job processing, which helped me understand how scalable and production-level systems are built.

My contribution to the PlayEvent client platform provided experience in working on real client requirements, improving backend systems, integrating frontend and backend components, and testing production-level applications. In addition, exploring modern technologies such as Bun, Hono, Drizzle ORM, Convex, and MCP helped me understand modern backend architectures and emerging development practices.

This internship significantly improved my understanding of scalable system design, production-level architecture, and real-world software engineering practices. I gained experience working with structured development workflows, version control systems, and real production codebases. Working in a professional environment helped me develop problem-solving skills, improve code quality, and understand how large-scale applications are designed and maintained.

Overall, the Practice School II internship at Eurobliz Technologies significantly improved my technical skills, practical knowledge, and understanding of modern software engineering. It provided strong exposure to real-world full-stack development and prepared me for future roles in software development and scalable SaaS application engineering.

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